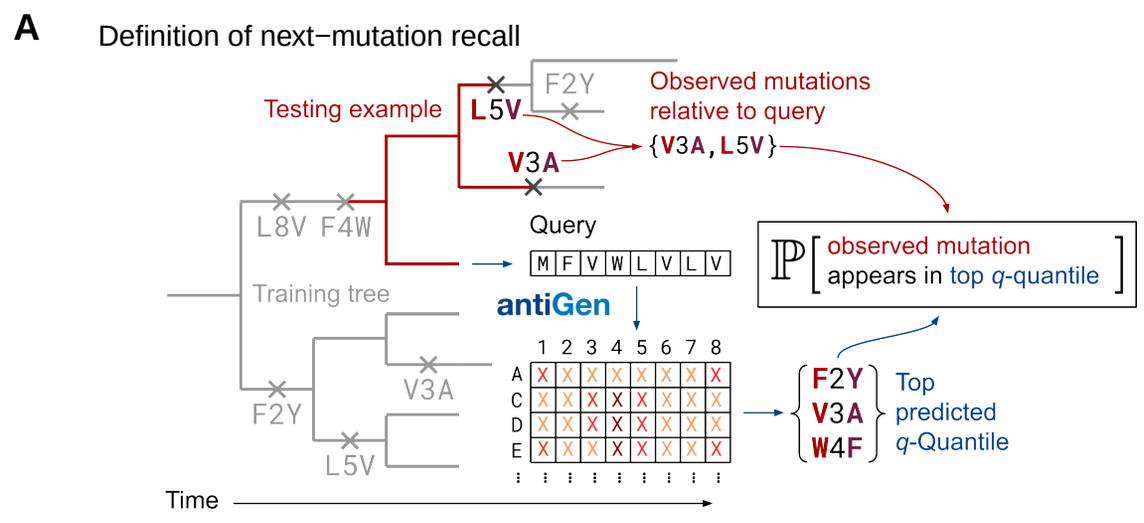
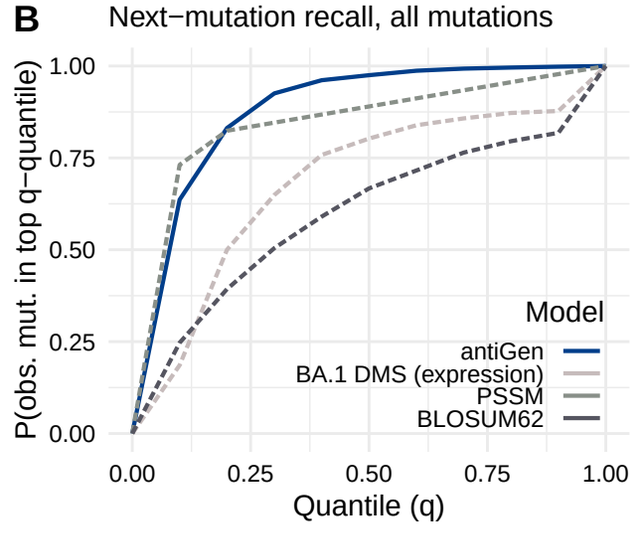
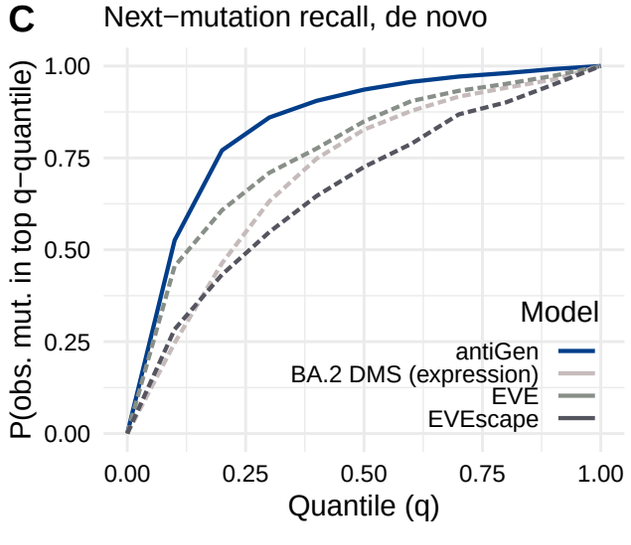
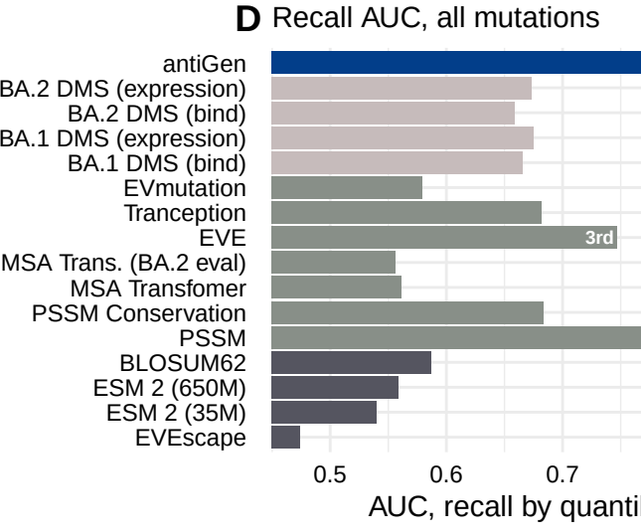
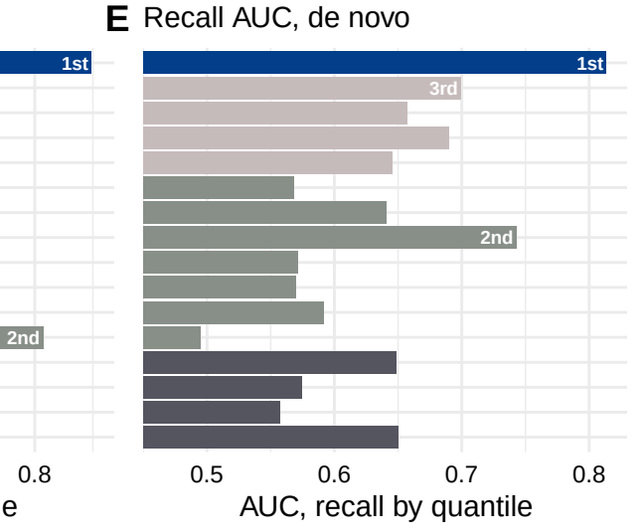
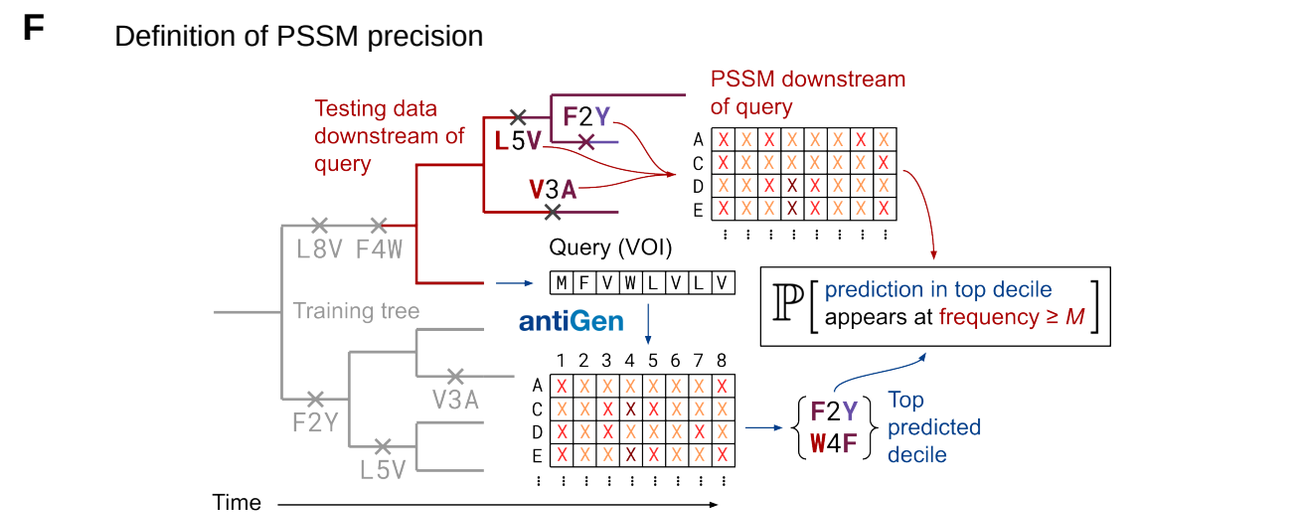

A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Query
M F V W L V L V
antiGen
Training tree
V3A
F2Y
L5V
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[observed mutation appears in top q-quantile]
F2Y
V3A
W4F
Top predicted q-Quantile
Time

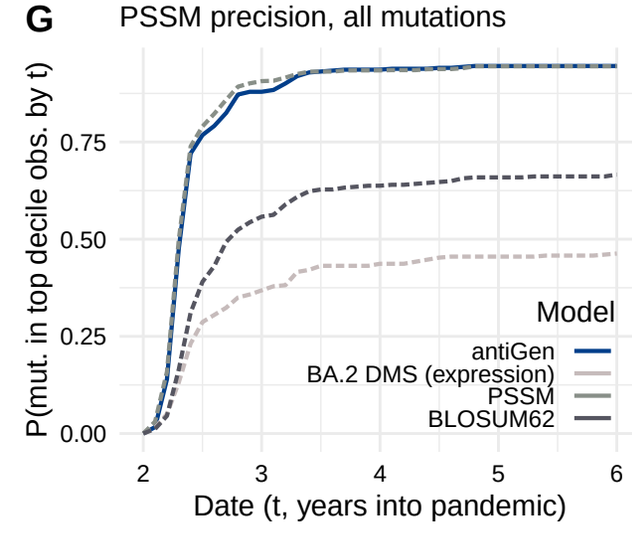

B
Next−mutation recall, all mutations
P(obs. mut. in top q−quantile)
1.00
0.75
0.50
0.25
0.00
0.00 0.25 0.50 0.75 1.00
Quantile (q)
Model
antiGen
BA.1 DMS (expression)
PSSM
BLOSUM62

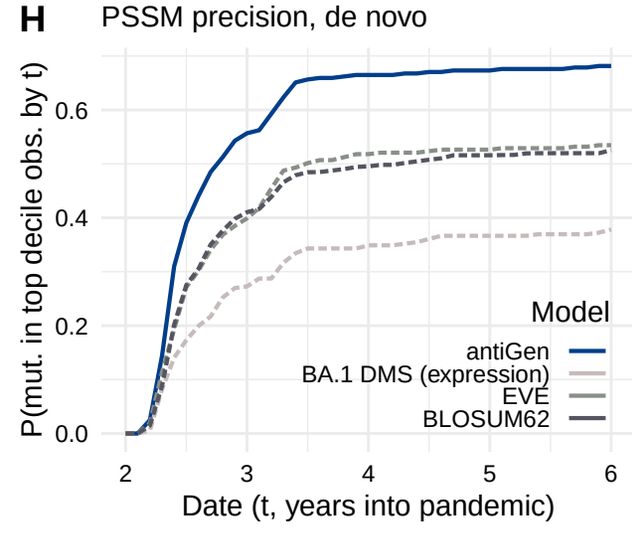

C
Next−mutation recall, de novo
P(obs. mut. in top q−quantile)
1.00
0.75
0.50
0.25
0.00
0.00 0.25 0.50 0.75 1.00
Quantile (q)
Model
antiGen
BA.2 DMS (expression)
EVE
EVEscape

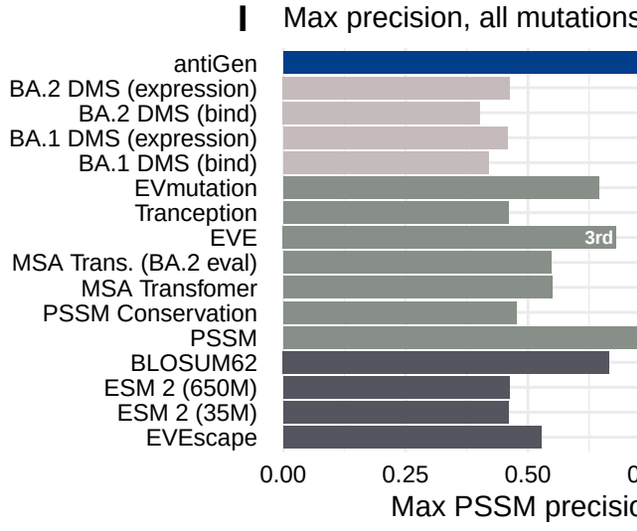

D
Recall AUC, all mutations
antiGen
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
EVmutation
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
1st
3rd
2nd
0.5 0.6 0.7 0.8
AUC, recall by quantile

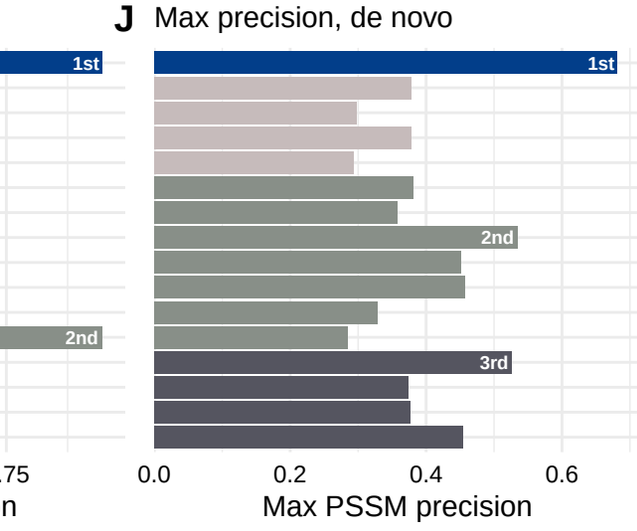

E
Recall AUC, de novo
1st
3rd
2nd
0.5 0.6 0.7 0.8
AUC, recall by quantile


F
Definition of PSSM precision
Testing data downstream of query
F2Y
L5V
V3A
PSSM downstream of query
L8V F4W
Query (VOI)
M F V W L V L V
antiGen
Training tree
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[prediction in top decile appears at frequency ≥ M]
F2Y
W4F
Top predicted decile
Time

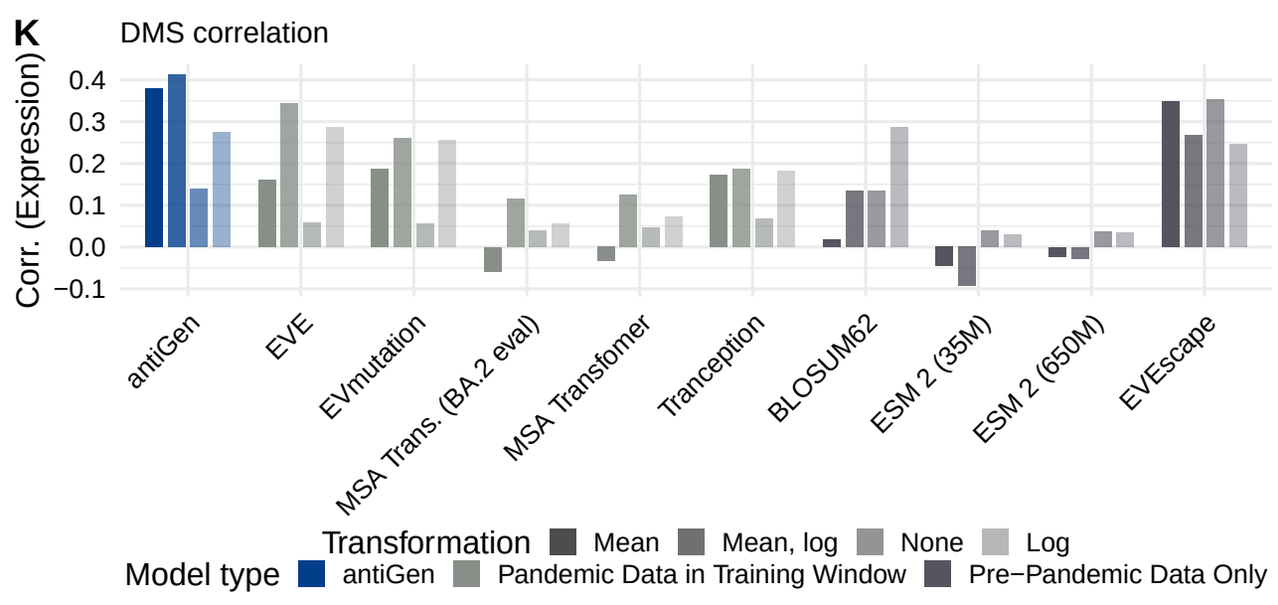

G
PSSM precision, all mutations
P(mut. in top decile obs. by t)
0.75
0.50
0.25
0.00
2 3 4 5 6
Date (t, years into pandemic)
Model
antiGen
BA.2 DMS (expression)
PSSM
BLOSUM62

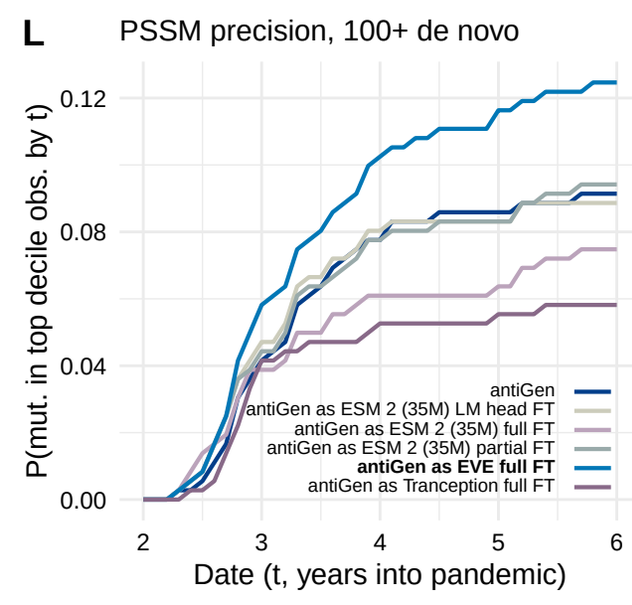

H
PSSM precision, de novo
P(mut. in top decile obs. by t)
0.6
0.4
0.2
0.0
2 3 4 5 6
Date (t, years into pandemic)
Model
antiGen
BA.1 DMS (expression)
EVE
BLOSUM62

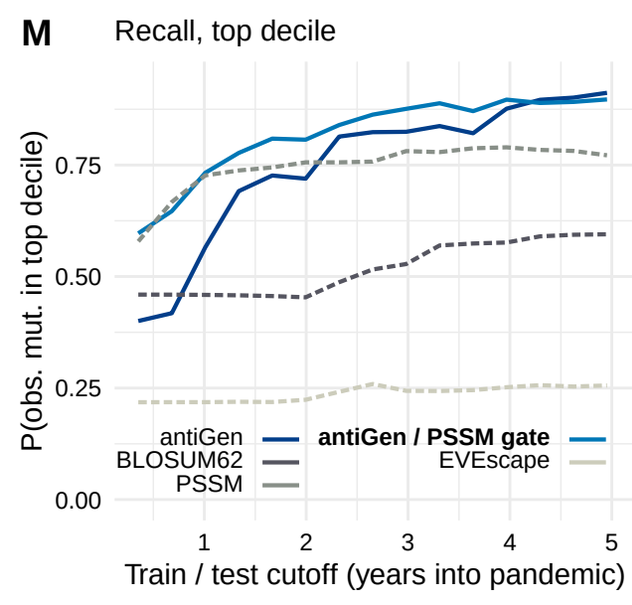

I
Max precision, all mutations
antiGen
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
EVmutation
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
1st
3rd
2nd
0.00 0.25 0.50 0.75
Max PSSM precision


J
Max precision, de novo
1st
2nd
3rd
0.0 0.2 0.4 0.6
Max PSSM precision


Model type
antiGen
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only


K
DMS correlation
Corr. (Expression)
0.4
0.3
0.2
0.1
0.0
−0.1
antiGen
EVE
EVmutation
MSA Trans. (BA.2 eval)
MSA Transfomer
Tranception
BLOSUM62
ESM 2 (35M)
ESM 2 (650M)
EVEscape
Transformation
Mean
Mean, log
None
Log
Model type
antiGen
Pandemic Data in Training Window
Pre−Pandemic Data Only


L
PSSM precision, 100+ de novo
P(mut. in top decile obs. by t)
0.12
0.08
0.04
0.00
2 3 4 5 6
Date (t, years into pandemic)
antiGen
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) partial FT
antiGen as EVE full FT
antiGen as Tranception full FT


M
Recall, top decile
P(obs. mut. in top decile)
0.75
0.50
0.25
0.00
1 2 3 4 5
Train / test cutoff (years into pandemic)
antiGen
BLOSUM62
PSSM
antiGen / PSSM gate
EVEscape